

AYUSH KUMAR

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[Portfolio](#) ◊ [GitHub](#) ◊ [GitLab](#) ◊ [LinkedIn](#)

SUMMARY

Computer Science student building backend, real-time, and AI-integrated systems with Python, TypeScript, FastAPI, Redis, Docker, and PostgreSQL. Work spans distributed simulation, AI memory infrastructure, trading workflows, and open-source contributions with merged changes in developer tooling and API behavior.

EDUCATION

RV Institute of Technology and Management

Bachelor of Engineering, Computer Science & Engineering

Expected May 2027

CGPA: 9.03 / 10.00, through 5th Semester

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, Go

Backend: FastAPI, Node.js, Express, REST APIs, WebSockets, SSE, async processing, Pydantic

Data & Systems: PostgreSQL, Redis, MongoDB, MySQL, pgvector, caching, indexed lookup, distributed systems

AI / ML: PyTorch, XGBoost, scikit-learn, HuggingFace Transformers, LangChain, NumPy, Pandas

Cloud & Tooling: Docker, Linux, Git, GitHub Actions, CI/CD, AWS, GCP, debugging, monitoring

Core CS: Data structures, algorithms, operating systems, databases, software design

EXPERIENCE

Open Source Software Contributor

2023 – Present

56 public contributions, 23 merged: 44 GitHub PRs and 12 GitLab MRs across Vitest, Pydantic AI, OpenTelemetry, WordPress, pyenv, NASA F Prime GDS, Boeing, ScanCode, OWASP, Processing, Google Gemini CLI, and CDLI Framework.

- Fixed tests, CLI workflows, API behavior, UI components, documentation, and developer tooling using regression tests and CI logs.
- Scoped changes from issue context and maintainer feedback, keeping patches small, reviewable, and compatible with existing project conventions.

PROJECTS

SENTINEL - Distributed Market Microstructure Simulator

Ongoing

Python, FastAPI, TypeScript, Next.js, WebSockets, XGBoost, scikit-learn

[Live Demo](#) | [GitHub](#)

- Built a distributed simulator with **40+ concurrent agents**, live order-book analytics, WebSocket telemetry, and risk/PnL dashboards.
- Engineered 6 trader types with latency injection, synchronization, stress parameters, and event-driven state updates.
- Built liquidity-shock prediction from spread, depth, imbalance, and volatility signals, reaching **85% detection accuracy** with **60–90s** lead time.
- Implemented TWAP/VWAP/Iceberg detection with **83% precision**; presented SENTINEL at IEEE ICWITE 2026.

ENGRAM - Self-Hostable AI Memory Layer for Cross-LLM Context Sync

May 2026

Python, FastAPI, TypeScript, PostgreSQL, pgvector, Docker Compose

[Live Demo](#) | [GitHub](#)

- Architected AI memory middleware with async extraction, deduplication, semantic search, confidence scoring, logs, and access tracking.
- Designed a FastAPI proxy for concurrent LLM requests with per-user retrieval, relevance ranking, and provider-neutral forwarding.
- Built a TypeScript MCP server and dashboard for memory inspection, semantic search, log analysis, and integration debugging.

EQUITYFLOW - Real-Time Paper Trading Platform

Feb 2026 – Mar 2026

Python, FastAPI, TypeScript, Next.js, SSE, REST APIs

[Live Demo](#) | [GitHub](#)

- Built broker-data, order, portfolio, cash, and P&L workflows with sub-second SSE updates and indexed contract resolution across **40,000+ instruments**.
- Added cache TTL controls, polling fallbacks, provider switching, stale-data detection, reconciliation, and stream diagnostics.

OCCASIO - Event Management and Booking Platform

Oct 2025 – Nov 2025

Node.js, Express, PostgreSQL, Prisma, Redis, BullMQ

[Live Demo](#) | [GitHub](#)

- Designed REST APIs for auth, payments, wallets, tickets, analytics, and teams with PostgreSQL workflows, Redis queues, QR verification, and payment integrations.

ACHIEVEMENTS & SIGNALS

Competitive programming: solved **300+ algorithmic problems** across Codeforces, LeetCode, CodeChef, and AtCoder; ranked **top 20% globally** in a **52,000+ participant** Codeforces contest. **Research and leadership:** presented SENTINEL at IEEE ICWITE 2026; published two Zenodo ML preprints; Technical Head, Robotics / Web Dev / Cybersecurity / Coding and Entrepreneurship Club.